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The interaction of flame and flow field in a lean premixed swirl-stabilized combustor operated on H_2/CH_4/air

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Abstract

The interaction of flame and flow field was studied in an enclosed lean premixed swirl-stabilized combustor operated on methane (CH_4)- and hydrogen (H_2)- enriched CH_4. This was accomplished by simultaneous measurements of the 2D velocity field using particle image velocimetry and planar laser induced fluorescence imaging of the hydroxyl (OH) radical. Experiments were conducted at an inlet Reynolds number of 13,500 and a theoretical swirl number of 1.5. The fuel was pure CH_4 or a mixture of 60% CH_4 and 40% H_2 by volume. The adiabatic flame temperature was 1350 ± 20 °C for both fuels. Results show that the addition of H_2 changes the shape of the reaction zone, producing a shorter, and a more robust flame. Fuel composition also affected the location of the reaction zone, which in turn, influenced both instantaneous and time-averaged flow fields. Time-resolved measurements show vortical structures containing regions of high OH concentration around their edges. The combustion stabilized in regions of high compressive strain for the CH_4 flame but not for the H_2-enriched CH_4 flame. In the CH_4 flame, reactions extinguished in the high velocity inlet jet region, where the H_2-enriched CH_4 flame was able to sustain combustion.
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Keywords: Gas turbine engines; Premixed turbulent flames; Laser- induced fluorescence

1. Introduction

Lean premixed (LPM) combustion is currently the method of choice for reducing pollutant emissions in land-based gas turbine engines. The lower peak flame temperatures in LPM combustion abate the formation of oxides of nitrogen (NO_x) via the thermal pathway. Most of the current LPM systems are limited to utilizing natural gas. However, gasified coal and biomass, chemical and refinery waste, and full-cell off-gases are some of the alternate fuel stocks that may potentially be used in gas turbines. Understanding the role of hydrogen (H_2), typically present in these fuels, is important to the larger goal of utilizing alternate fuels for power generation. Hydrogen-enriched hydrocarbon fuels may also provide a cross-over strategy to the H_2 energy infrastructure of the future.

Past experiments have shown improvement in flame stability and/or NO_x reductions when H_2 was added to the hydrocarbon fuel [1–3]. Ren

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et al. [4] used an opposed-jet, symmetric, twin-flame configuration to find that the presence of H_2 in methane (CH_4) resulted in a substantial increase of strain rate at extinction, especially at lower equivalence ratios. Similar results were obtained by Jackson et al. [5], who found early H_2 breakdown increased radical production near the base of the flame and thereby, increased the burning rates. Schefer et al. [6] conducted experiments in a swirl-stabilized combustor using CH_4 and H_2-enriched CH_4. Results indicated improved flame stability and extended lean blow-off (LBO) limit with H_2 addition. Moreover, the visual flame shape and OH levels were also affected by the fuel composition.

In recent years, greater insight into the reacting flow field has been gained by utilizing particle image velocimetry (PIV). Ji and Gore [7] measured the flow field in a swirl-stabilized, unconfined burner and found significant differences between mean and instantaneous flow structures. For example, the mean flow exhibited a large vortex, while several small-scale vortices were observed in the time-resolved flow field. Similar results were obtained by Wicksall et al. [8] in confined swirling flows, with and without chemical reactions. Heat release in the flame was found to affect both mean and instantaneous flow structures, indicating strong coupling between fluid flow and chemical reactions. Results indicated that the information about the interaction between reaction zone and flow field is necessary to understand and to predict the combustion process.

Several research groups have obtained simultaneous measurements of velocity and scalar fields using PIV and planar laser induced fluorescence (PLIF) imaging of hydroxyl (OH) or methyl (CH) radical [9–14]. Rehm and Clemens [10] found that thin OH zones are associated with regions of high compressive strain in non-premixed hydrogen jet flames. Donbar et al. [11] found highly wrinkled, but rarely broken CH layer in turbulent non-premixed flames. Data revealed strain rates of up to $20{,}000\,s^{-1}$ at the tip of the flame. Measurements by Kothnur et al. [13] show that CH structures exhibit a stronger correlation with changes in the flow field resulting from heat release than OH structures. Han and Mungal [14] found that intense combustion occurred in regions of high compressive strain.

The present study seeks to understand the interaction between reaction zone and turbulent flow field. Unlike past studies of non-premixed flames, we consider LPM combustion typically employed in gas turbine engines. Swirl-stabilized combustion occurring inside a chamber is investigated to represent a widely used industrial flow configuration. The operating variable is the fuel composition, i.e., pure CH_4 or H_2-enriched CH_4, to emphasize the role of the chemistry. Simultaneous measurements of the flow field and OH radical are obtained, respectively, using PIV and OH-PLIF imaging. The rest of the paper will consist of a description of the experimental apparatus, the results obtained, a brief discussion, and concluding remarks.

2. Experimental apparatus

Experiments at atmospheric pressure and temperature were conducted in an enclosed swirl-stabilized LPM combustor described by Wicksall et al. [8]. The fuel was injected into the air 400 mm upstream of the combustor. The entrance to the combustor contained a six-vane axial swirler located in the annulus around the centerbody. The vanes were oriented at a 28° angle to the combustor inlet plane. The diameter of the centerbody was 20 mm and the outer diameter of the swirler was 41 mm. The combustion chamber was a 300 mm long quartz glass tube with an inner diameter of 83 mm and a 4 mm thick wall, which was cooled by natural convection. Half of the inside of the quartz tube was sand blasted to reduce reflection of the laser light. Experiments were conducted at an inlet Reynolds number of 13,500, which corresponds to a bulk axial inlet velocity of 10 m/s. The Reynolds number was calculated using the bulk axial velocity, the hydraulic diameter of the annulus around the centerbody, and the properties of air. The theoretical swirl number of 1.5 was calculated assuming that the flow leaves the swirler parallel to the vanes. The fuel was 99% pure CH_4 or a mixture of 60% CH_4 and 40% H_2 by volume. The adiabatic flame temperature was 1350 ± 20 °C for both fuel mixtures. This flame temperature corresponds to an equivalence ratio of 0.58, a value slightly higher than the lean stability limit of the CH_4 flame. These conditions pertained to typical non-sooting blue flames that were visibly stable. Additional physical description of these flames is provided by Schefer et al. [6].

A dual-chamber Nd:YAG laser was used as the illumination source in the PIV system. The laser sheet was 50 mm high and 0.5 mm thick, and the time delay between pulses was 15 μs. The strong out-of-plane motion caused by the swirling flow necessitated shortening the time delay and increasing the thickness of the laser sheet to avoid excessive loss of particles out of the measurement plane [7]. The laser light scattered off the aluminum oxide seed particles, and was recorded on consecutive frames of a 1024×1024 pixels digital camera with a field-of-view of 30×30 mm. The PIV images were analyzed using PIV LAB 2000 (version 1.50) Matlab Routines developed by Han and Mungal [15]. The interrogation region was 32×32 pixels, or 0.9 mm per side. A 50% overlap of interrogation regions was used, resulting in velocity vectors at every 0.45 mm. Further details of the PIV system used in this experiment

are given by Wicksall et al. [8]. The uncertainty in the velocity calculated using the methodology proposed by Haste [16] was 0.31 m/s.

A Q-switched Nd:YAG laser operating at 532 nm was used to pump the dye laser in the OH PLIF system. A beam at 283.556 nm was produced by frequency doubling the output of the dye laser. This frequency excited the Q1(8) line of the (1,0) band of the $A^2\Sigma \leftarrow X^2\Pi$ electronic transition. The Q1(8) transition was utilized because the signal was proportional to the mole fraction within 10% of the peak value, without the need for population fraction and quenching corrections [17]. The sheet forming optics utilized for the PIV beams were also used for the PLIF system. The PLIF laser pulse was coplanar with the two PIV pulses and occurred between them. The PIV and OH signals were collected with a single lens and separated using a dichroic placed at 45° angle to the laser sheet, which turned the OH signal perpendicular to the PIV signal, while the PIV signal passed through the dichroic unaffected. Further details of the OH PLIF system used in this experiment are given by Schefer et al. [6].

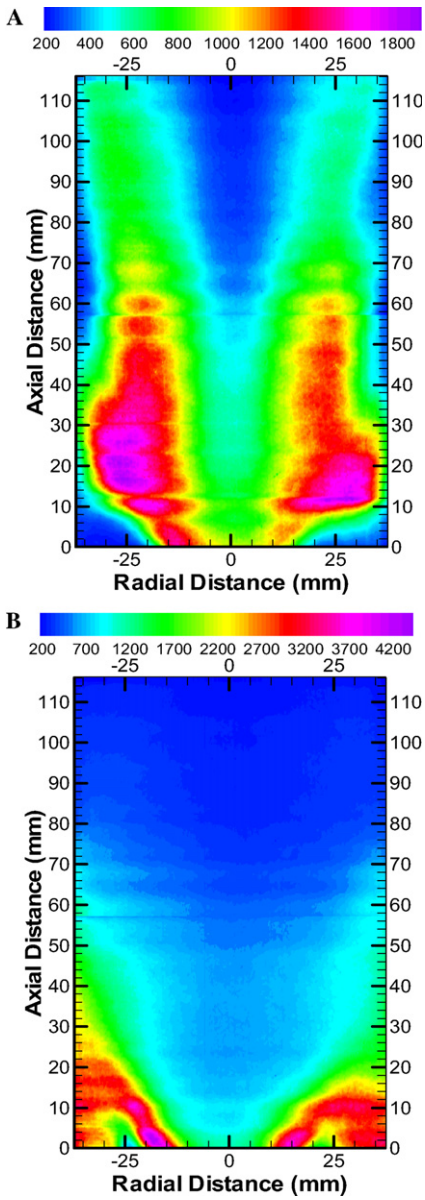

Fig. 1. Time-averaged OH PLIF image for (A) CH_4 flame, (B) H_2-enriched flame.

3. Results and discussion

In this section, time-averaged data are presented followed by the simultaneous measure-

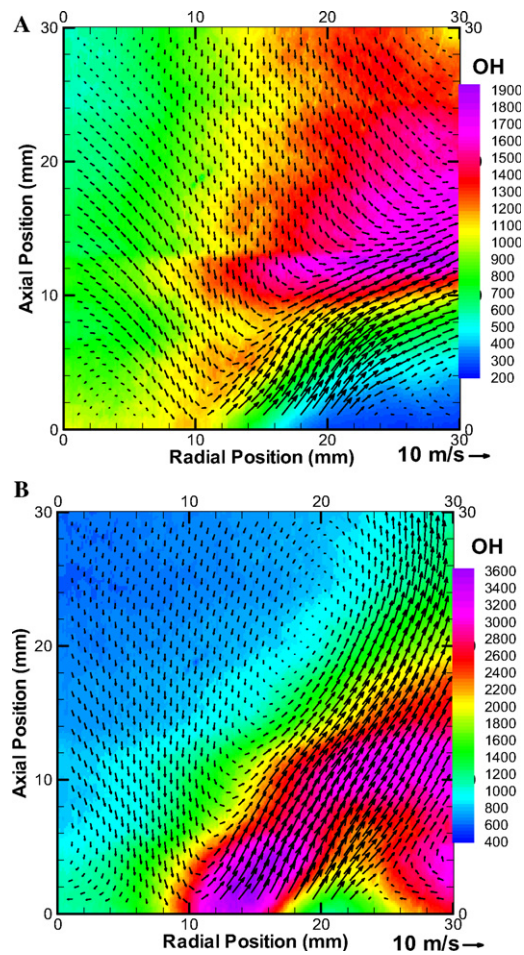

Fig. 2. Time-averaged OH PLIF image superimposed with velocity vectors (only 11% of the vectors are shown) for (A) CH_4 flame, (B) H_2-enriched flame.

ments. The measurements were taken in many separate trials as only 10 data sets could be obtained at a time because of fouling of the quartz combustor by the seed particles. The time-averaged velocity data were computed from a sequence of 200 instantaneous measurements, which were indistinguishable from averages performed with 100 data points. The averaged OH data presented are from a set of 1000 images.

3.1. Time-averaged measurements

The overall understanding of the flame structure is gained from Fig. 1, showing time-averaged OH images for CH_4 and H_2-enriched CH_4 flames. The horizontal lines seen in the OH image are due to distortion by the quartz confinement tube. The standard corrections for flat field, background, and laser sheet non-uniformity have been made to all the OH images presented in this work. In Fig. 1, a different color scale is used for each flame. The OH image for the CH_4 flame (Fig. 1A) shows long and wide reaction zones on either side of the jet axis. Reactions occur mainly up to the axial location, $z = 60$ mm, although minor quantities of OH persist farther downstream. No OH is found at the intersection of the combustor inlet plane and wall, indicating a lack of reactions in the corner recirculation zone. In contrast, the time-averaged OH image for the H_2-enriched flame is shorter and more intense, with higher OH levels. The combustion is virtually complete at $z = 25$ mm. High OH levels in the corner recirculation zone demonstrate intense combustion in that region. Minor asymmetry in Fig. 1 is attributed, in part, to the differences in velocity profiles at the leading and trailing edges of the inlet swirler vanes.

Although OH data in Fig. 1 were obtained across the full length of the combustor, simultaneous data were acquired on only one side of the jet axis over the 30×30 mm field-of-view of the PIV setup. Figure 2 shows the time-averaged OH PLIF data overlaid with velocity vectors for both flames. The flow field can be divided into several major regions. The inlet jet denoted by the high velocity reactant inflow is located at radial positions between 10 and 20 mm. The large backflow in the center of the combustor is the centerbody recirculation region. The corner recirculation zone (in the lower right side of the image) is formed at the intersection of the combustion inlet plane and wall. The inner shear layer is located between the centerbody recirculation zone and the inlet jet, and an outer shear layer resides between the inlet jet and the corner recirculation region.

The time-averaged CH_4 flame data in Fig. 2A show high OH levels in the inner shear layer near the combustor inlet and farther downstream. Combustion is not sustained in the high velocity region of the inlet jet. Hot combustion products in the inner shear layer and centerbody recirculation region are shielded by the inlet jet from reaching into the outer shear layer and the corner recirculation region and hence, no reactions are occurring in these zones. In contrast, Fig. 2B shows combustion occurring in the inlet jet region for the H_2-enriched flame. Consequently, the outer shear layer and corner recirculation region also participate in combustion to help produce a shorter flame. Consumption of fuel in the corner recirculation zone and the resulting products, mixing with the incoming reactants would also enhance the reaction rates. The reduction in flame length, however, may not be entirely due to the added combustion in the recirculation zone. Fuel-composition-dependent location of heat release has also affected the global features of the flow field. The inlet jet in the H_2-enriched flame has a higher momentum because of the heat release occurring within it. The increased velocity causes the inlet jet region to align more vertically to a 60° angle with respect to the combustor inlet plane. In the CH_4 flame, the non-reacting inlet jet region lacks momentum and hence, the jet tilts towards the wall upon entering the combustor. The OH image in Fig. 2B does not show correlation with the velocity field at $z > 20$ mm because there is no reaction zone at downstream locations as shown in the image. The low OH levels at these locations correspond to OH molecules persisting because of the slow third body reactions.

Figure 3 shows profiles of the time-averaged axial velocity, OH, 2D vorticity, and 2D normal strain near the combustor inlet ($z = 2.5$ mm) for both flames. The vorticity and normal strain were calculated using the path integral method to avoid differentiation errors associated with correlated data in an over-sampled PIV field [18]. The uncertainty in the velocity measurements has resulted in an uncertainty in the vorticity and strain fields of $400\ s^{-1}$ [18]. Figure 3A shows the mean axial velocity and OH signal for the CH_4 flame at axial plane $z = 2.5$ mm. The velocity profile shows the centerbody recirculation zone in back flow, the inner shear layer where the axial velocity changes sign, the inlet jet with its high velocities, and the outer shear layer. The peak OH is observed in the inner shear layer, where the flame stabilizes. Figure 3B shows vorticity and normal strain profiles at the same axial location as Fig. 3A. A positive vorticity peak is observed in the inner shear layer. The peak vorticity location coincides with the location of the negative peak normal strain. The negative normal strain indicates compression of fluid elements in the inner shear layer. Results show that combustion in the CH_4 flame is stabilized in regions of high compressive strain, which agrees with similar observations made by others in non-premixed flames [10,14]. Figure 3C shows the axial velocity and OH profiles for the H_2-enriched flame. Interestingly, the OH peak has shifted into

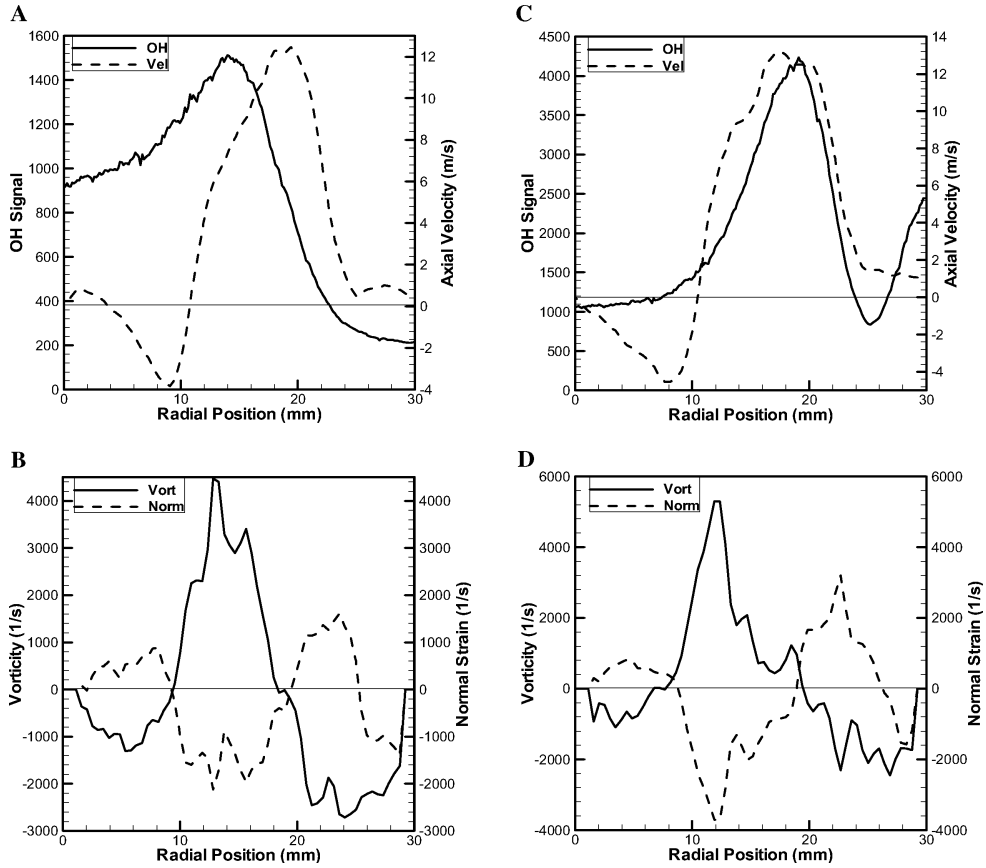

Fig. 3. Time-averaged radial profiles at $z = 2.5$ mm. (A) and (B) CH_4 flame, (C) and (D) H_2-enriched flame.

the high velocity inlet jet region. The ability of the flame to stabilize in the high velocity jet region is attributed to H_2's ability to withstand higher strain before extinction, as compared to CH_4. Furthermore, heat and combustion products from the inlet jet help sustain combustion in the corner recirculation zone for the H_2-enriched flame. Figure 3D shows vorticity and normal strain profiles for the H_2-enriched flame. Again, we see a vorticity peak associated with the inner shear layer coinciding with the region of strong compression but not the reaction zone. These results indicate that chemical aspects take precedence over flow aspects in determining the location of heat release in the H_2-enriched flame.

3.2. Time-resolved measurements

Figure 4 shows the simultaneous OH and velocity data at four random instants in time for the CH_4 flame. The OH fields show significantly different flame structures, and the velocity fields show many small vortical structures that were not observed in the time-averaged field. In Fig. 4A, the inlet jet region is highly erratic and without combustion. A vortical structure is observed downstream of the inlet jet at radial position, $r > 20$ mm and $8.0 < z < 15.0$ mm. A thin OH layer curves around the left (inner) side of the vortical structure such that the OH gradient is nearly perpendicular to the flow direction. Figure 4B shows a similar vortical structure centered at $r = 20$ mm and $z = 17$ mm. Again, the OH layer is observed to wrap around the left (inner) side of the vortex. These vortical structures may be 2D projections of 3D rotating filaments with combustion occurring around them. Evidently, the vortical structures are important to flame stabilization because they provide a source of heat and radicals to the incoming reactant stream. Combustion is observed in the outer edges of these structures where there is mixing between the vortex and its surroundings, but not in the interior where there is no supply of reactants. Figure 4C shows a significantly different flow field and flame structure compared to that at other instants. A very strong backflow with high OH levels is located around $r = 10$ mm and $z = 10$ mm. The backflow is expected to carry combustion products to burn fresh reactants along its path.

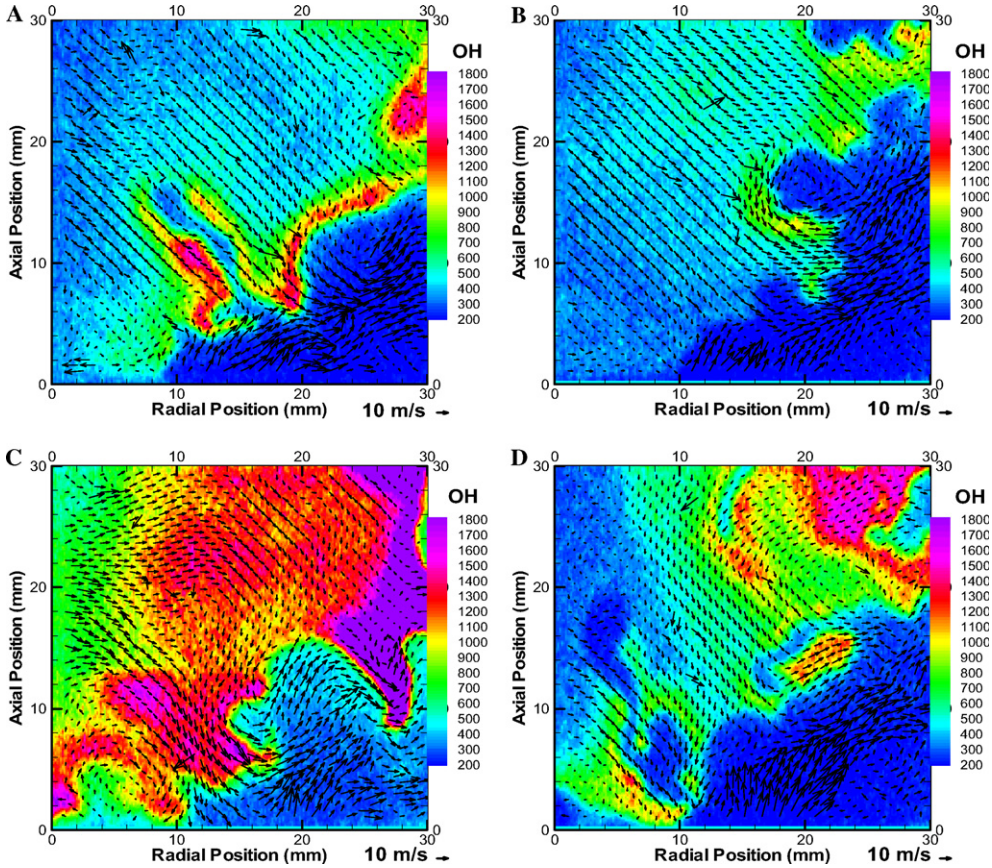

Fig. 4. OH PLIF image superimposed with velocity vectors (only 11% of the vectors are shown) at four random instants of time for the CH_4 flame.

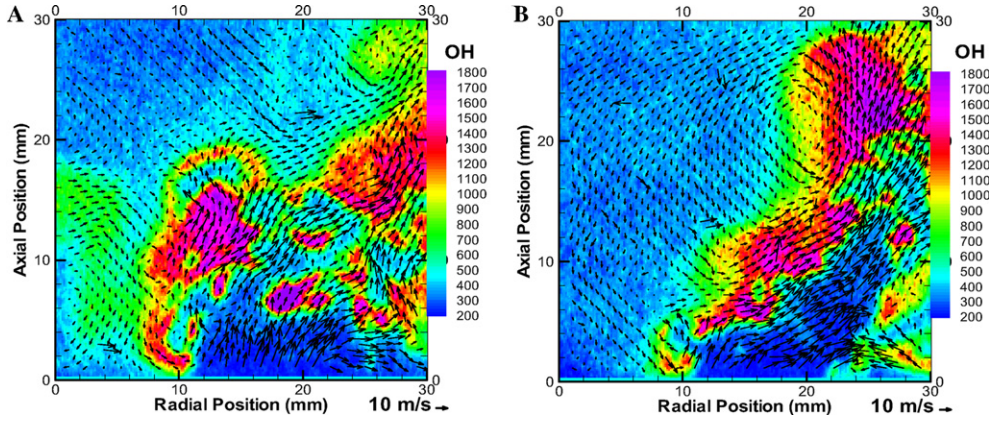

Fig. 5. OH PLIF image superimposed with velocity vectors (only 11% of the vectors are shown) at two random instants of time for the H_2-enriched flame.

A large vortex with a surrounding OH layer is observed at the jet center near the combustor inlet. Another feature is the filaments of OH extending downwards along the periphery of the combustor. The flow field is observed to contain vortical structures around the filaments and small flow velocities within the filament, possibly explaining the stabilization of combustion. Figure 4D con-

tains a large vortex at $r = 25$ mm and $z = 20$ mm with much of the OH around it. Again, vortices appear to help stabilize combustion in some instances.

Figure 5 shows simultaneous OH and velocity data for the H_2-enriched flame. Data are shown only for two instants because flame and flow structures at other instants were similar to one another. The OH field in Fig. 5A shows many small-scale structures with intense combustion, compared to the larger structures observed in the CH_4 flames. The CH_4 flame was closer to extinction and therefore, it could not sustain combustion at small scales without adequate flame stabilization. A small vortex with an OH layer around it is seen near the combustor inlet at $r = 10$ mm, and at other locations in the combustor. One interesting observation is that the flame also stabilizes in regions without flow recirculation. For example, strong upward flow coinciding with high OH is observed at $r = 10$ mm and $z = 15$ mm, and $r = 25$ mm and $z = 15$ mm. Figure 5B contains a large vortical structure at $r = 20$ mm and $z = 20$ mm. The right (outer) side of this vortex overlaps a significant portion of the OH region, and the left (inner) side has virtually no OH. This is explained by the combustion occurring upstream in the inlet jet, which provides hot products to sustain combustion on the right side of the vortex. Regions of OH in Fig. 5 indicate that reactions in the corner recirculation zone and the outer shear layer are assisted by combustion occurring in the inlet jet.

Figure 6 shows profiles of instantaneous axial velocity, OH, 2D vorticity, and 2D normal strain at $z = 10$ mm derived from data in Figs. 4A and 5A. Figure 6A shows profiles of axial velocity and OH for the CH_4 flame. The velocity profile shows a large central recirculation region, the inner shear layer, and the inlet jet. A large OH peak is observed in the central recirculation region. Figure 6B shows the vorticity and normal strain for the above case. There appears some correlation between vorticity and normal strain, as spikes in vorticity often occur near spikes in normal strain, although the correlation in not perfect. Figure 6A and B indicate that the peak OH occurs near the location of peak compressive strain, as observed previously from the time-averaged data. Figure 6C shows profiles of axial velocity and OH for

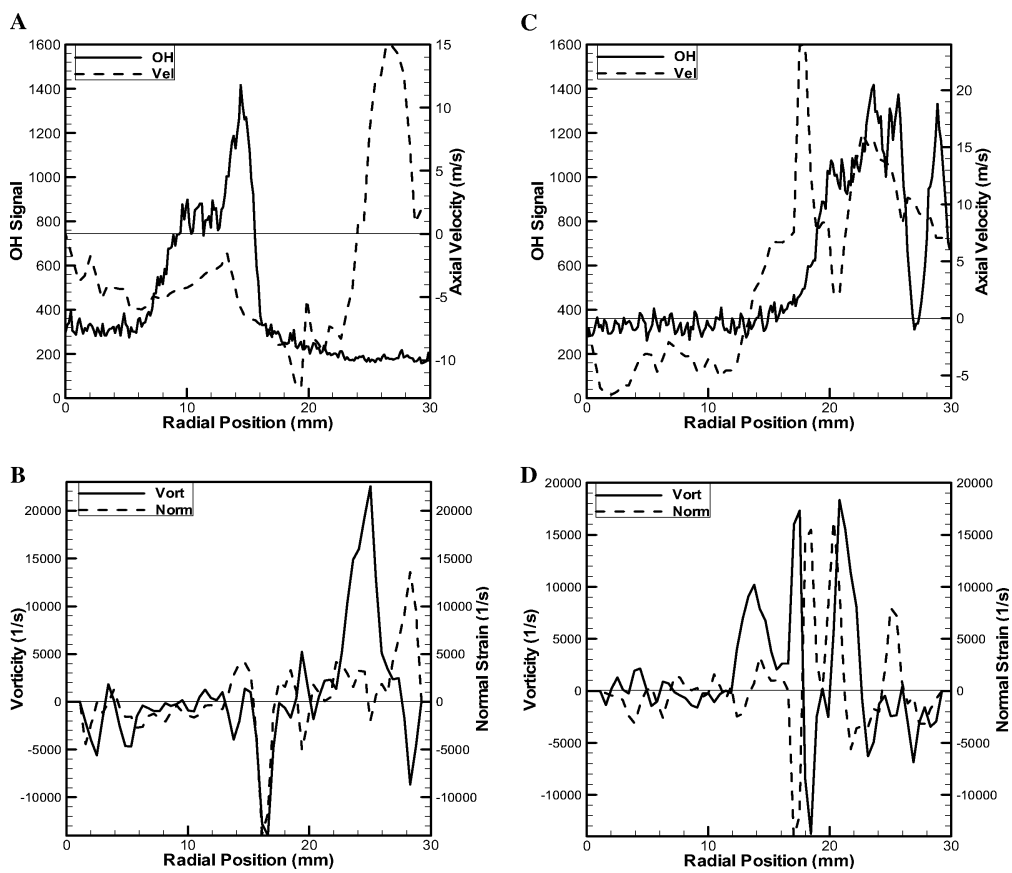

Fig. 6. Instantaneous radial profiles at $z = 10.0$ mm. (A) and (B) CH_4 flame, (C) and (D) H_2-enriched flame.

the H_2-enriched flame. The location of the peak OH has shifted towards the inlet jet region. An OH peak is also observed in the corner recirculation region. The differences in the location of the reaction zone are explained by enhanced stability of the H_2-enriched fuel. Figure 6D shows that the correlation between vorticity and OH level is not observed for the H_2-enriched flame. Likewise, there is lack of correlation between the OH level and normal strain.

4. Conclusions

Simultaneous measurements of OH and velocity fields were obtained in an enclosed lean premixed swirl-stabilized combustor operated on CH_4 and H_2-enriched CH_4. Results show that the flame and flow structures for the two cases are different. The CH_4 flame was longer, and lacked combustion in the corner recirculation region, whereas the H_2-enriched flame was shorter and more robust with reactions occurring in the corner recirculation region. Reactions in the CH_4 flame stabilized in the inner shear layer whereas those in the H_2-enriched flame stabilized in the inlet jet region. The differences in the location of heat release for the two flames will affect important combustion characteristics such as the pattern factor and wall heat transfer. The time-averaged flow fields also differed, in that the reacting inlet jet of the H_2-enriched flame maintained its orientation farther downstream while the non-reacting inlet jet of the CH_4 flame lost its momentum and tilted towards the combustor wall. The simultaneous images show several small-scale vortical structures wrapping OH layers around them, which was not visible in the time-averaged fields. The H_2-enriched flame also stabilized in regions without flow recirculation. Reactions in the CH_4 flame stabilized at the location of the highest compressive strain. However, reaction zones of the H_2-enriched CH_4 flame were less influenced by the flow field, indicating H_2's high resistance to strain providing improved flame stability.

Comments

Bala Varatharajan, General Electric Global Research, USA. Do you think your result will change for different flow velocities?

Reply. The trends observed are expected to be similar for different flow rates. However, the numerical values may change because flame properties are strain dependent, and strain scales with flow rate.

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Acknowledgment

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Matthew Juniper, University of Cambridge, UK. Did you look at varying the equivalence ratio and the hydrogen content simultaneously? Could a series of contours be plotted in the (Φ, $[H_2]$) space, along which the flame/flow field interaction is unchanged?

Reply. We did not vary equivalence ratio and hydrogen content simultaneously. It is conceivable that the flame/flow field interaction would remain unchanged at high equivalence ratios and high H_2 concentrations.

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Peter Jansohn, Peter Scherrer Institute, Switzerland. What made you choose a 60/40 = CH_4/H_2 mixture? Did you perform experiments at lower H_2 content and did you experience similar effects already at low H_2 content in the fuel mixture?

Reply. Fuel composition was chosen to represent moderate amounts of H_2 enrichment and to readily highlight the impact of fuel composition on flame/flow field interaction. We did not perform experiments with lower H_2 content.

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V.M. Rodriguez-Martinez, Universidad Carlos III, Madrid, Spain. Since the level of swirl is of high importance for the flame stabilization mechanism, I would appreciate the author's comments on:

1. The reason for a relatively high swirl number, Sg = 7.3, if any.
2. Whether they have analyzed lower swirl numbers, let's say $1 < Sg < 2$ and what the effects were/might be on flame shape, compactness, etc.?

Reply. The swirl number for experiment was 1.5; the value of 2.3 in the presentation was erroneous and has been corrected. We have not conducted a systematic study of the effects of swirl on simultaneous PIV/OH PLIF measurements. However, OH PLIF measurements performed at a lower swirl number of 0.78 have shown similar results ([6] in paper).

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Francesco Ciucci, Caltech, USA. I would like to understand the purpose of the quartz tube and how it affects your flame structure and your OH-LIF measurements. Is your OH-LIF signal affected by the scattering from the tube? And are the stripes I saw in your first slide regarding OH-LIF generated by the scattering due to the roughness of the tube? Furthermore, is the flame affected by the tube (change in boundary conditions), and if yes, in which measure? Regarding OH-LIF, did you correct your signal considering the non-homogeneous spatial laser power of your laser source? Also, what is the line-width of your laser (1 cm^{-1}) and what kind of impact does it have on your measurement?

Reply. Quartz tube was chosen to simulate a typical gas turbine combustor featuring enclosed swirl stabilized flame. The enclosure alleviates entrainment of surrounding air, constrains spreading of the jet/plume structures, and creates a corner recirculation zone at the intersection of the combustor inlet and wall. These changes introduced by the enclosure also affect the flame structure.

The stripes observed in the OH-PLIF images are possibly caused by non-uniformity of the quartz tube. Standard corrections including that for spatial non-homogeneity of the laser sheet were made. The line width of the YAG-pumped dye laser is about 0.08 cm^{-1}. The line width of the laser does not have any impact on the measurement. It is sufficiently small that we are only pumping a single OH line. Also, while the line width might affect absolute OH concentrations, we are accurately reproducing the spatial distribution and relative OH magnitude.